



## Calibrate crop growth

Estimate shared crop parameters  
Evaluate development and growth



## Calibrate crop water use

Fit irrigation-treatment responses  
Retrospectively test crop water use

## Simulate continuous regional rotations

Carry soil water across wheat, maize and fallows  
Compare crop responses at five irrigation levels



## Allocate irrigation spatially

Compare uniform and targeted allocation  
Hold the regional water budget constant  
Selection: 1997–2013



## Select annual quotas

Compare storage-rainfall and rainfall-only  
Match monitoring availability  
Selection: 2003–2013

## Evaluate management outcomes: 2014–2025

Rotation grain · irrigation · ET · drainage · spatial gains and losses